Higher National Diploma in Engineering (Mechanical/Civil)

Higher National Diploma in Building Service Engineering

First Year, First Semester Examination – 2018

CE-1102-Fluid Mechanics

Answer any five questions. **3hours**

Question 1

a) Define the following

i. Kinematic viscosity

ii. "Newton's law" with respect to viscosity

iii. Newtonian fluids

(10 marks)

b) The **velocity distribution** of a flow over a parabolic profile with a vertex **30 cm** from the plate, where the velocity is $180\ \mathrm{cm/s}$. If the viscosity of the fluid is $0.9\ \mathrm{Ns/m^2}$. Find the **velocity gradients** and the **shear stress** at distances of **0** and **15 cm** from the plate. Assume the equation of velocity profile for the parabolic is $u = ty^2 + my + n$. (10 marks)

Question 2 (Momentum Equation on a Bend)

a) **Derive an expression** for the magnitude and direction of the resultant force on the **horizontal reducing bend** which deflects the flow of fluid density ρ through an angle θ. Take at entry pressure P_1, area A_1, velocity V_1 and at exit pressure P_2, area A_2, velocity V_2 and the discharge being Q. (10 marks)

b) A **600 mm diameter main pipe** carries water under a pressure head of $25\ \mathrm{m}$ having flow velocity $2\ \mathrm{m/s}$. The main is fitted with a bend which turns the axis through 75°. Calculate the **resultant force** on the bend. Assume the bend is on the horizontal plane. (05 marks)

Question 3 (Hydrostatic Forces)

a) **Explain the terms** "**Total Pressure**" and "**Centre of Pressure**". (06 marks)

b) An **isosceles triangular plate** of base $5\ \mathrm{m}$ and altitude $5\ \mathrm{m}$ is immersed **vertically** in an oil of **specific gravity** 0.8. The base of the plate is $1\ \mathrm{m}$ below the free water surface, determine: I. The **total pressure**. II. The **center of pressure**. (14 marks)

Question 4 (Energy Equation and Head Loss)

a) In a vertical pipe carrying water, pressure gauges are inserted at points A and B where the pipe diameters are $0.3\ \mathrm{m}$ and $0.15\ \mathrm{m}$ respectively. The point B (inlet) is $1.5\ \mathrm{m}$ **below** A **(exit)** and when the flow rate down the pipe is $0.04\ \mathrm{m}^3/\mathrm{s}$, the pressure at B is $14500\ \mathrm{N/m}^2$ **greater than that at** A. Assuming the **losses** in the pipe between A and B can be expressed as $k(v^2/2g)$ where v is the velocity and g is the acceleration due to gravity at A, **find the value of** k. (10 marks)

b) Oil of density $750\ \mathrm{kg/m}^3$ flows in a pipe $80\ \mathrm{mm}$ **bore** with a mean velocity of $4\ \mathrm{m/s}$. The mean surface roughness is $0.016\ \mathrm{mm}$ and the length is $60\ \mathrm{m}$. The dynamic viscosity is $850\ \mathrm{kg/ms}$. I. **Determine the pressure loss**. (05 marks) II. **Calculate the minimum power required to drive a pump**. (05 marks)

Question 5 (Short Descriptions)

Write short description on the following topics:

Topic	Marks
(a) **Newtonian Fluid**	(2 marks)
(b) **Laminar Flow**	(2 marks)
(c) **Turbulent Flow**	(2 marks)
(d) **Boundary Layer**	(2 marks)
(e) **Francis Turbine**	(4 marks)
(f) **Kaplan Turbine**	(4 marks)
(g) **Pelton Wheel**	(4 marks)